

Thomas Liu

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EDUCATION

University of Waterloo

Sep. 2022 – Present

Bachelor of Mathematics - Major in Statistics and Computational Mathematics

Waterloo, ON

- Relevant Courses: Statistical Learning: Advanced Regression, Neural Networks, Computational Inference, Statistical Analysis, Forecasting, Applied Linear Models, Sampling & Experimental Design, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python, SQL, R, C, C++, JavaScript, LaTeX, C#, MATLAB, DAX

Data Tools: PowerBI, Excel, Snowflake, Git, AWS, Microsoft Azure, Looker/Data Studio, Google Analytics

Frameworks: Numpy, Pandas, Matplotlib, TensorFlow, PyTorch, Scikit-learn, XGBoost, PySpark, JSON

EXPERIENCE

Risk Management - Model Validation (Retail)

Sep. 2026 – Dec. 2026

TD Bank

Toronto, ON

Data Analyst

May 2026 – Aug. 2026

Rocket

Toronto, ON

- Engineered an automated **SQL** and **Python** mismatch detection pipeline for preferred vs. actual loan closing types, implementing automated email alerts to expedite critical error resolution
- Constructed interactive **PowerBI** dashboards to analyze internal and external AI usage, providing business stakeholders with key insights on AI's interaction with company data and closing workflows

Data Analyst

Sep. 2025 – Dec. 2025

Super.com

Toronto, ON

- Leveraged **SQL** and **Python** to benchmark QL2 against Luminai for competitor insights, confirming QL2's superior data volume and accuracy with **half the cost**, leading to a strategic supplier transition
- Led dbt model enhancements to incorporate new metasearch data sources, enabling robust cross-channel cMAB analysis and optimized data-driven strategy planning across travel acquisition channels
- Analyzed LLM-based room-matching results using **SQL** and collaborated closely with engineering to tune the model, resolving customer-reported mismatches and increasing overall revenue by **1.5%**

Data Analyst

Jan. 2025 – Apr. 2025

Lifestyle Home Products

Toronto, ON

- Built comprehensive, dynamic dashboards by integrating **Salesforce** data with **Google Sheets** and **Looker Studio**, enabling real-time insights that supported data-driven decision-making across business teams
- Established a stable **POST API** connection to securely transmit real-time customer data from **Salesforce** to third-party system Bellastone, ensuring reliable and efficient cross-platform data exchange
- Implemented automated data validation workflows within **Salesforce** to ensure up to **98%** data accuracy and completeness, enhancing reporting reliability, sales forecasting, and business intelligence processes

Data Analyst

May. 2024 – Aug. 2024

Ontario Lottery and Gaming Corporation

Toronto, ON

- Conducted post-campaign analysis using **Python** with **Numpy**, **Pandas**, and **Matplotlib**, identifying key audience segments and optimizing marketing strategies to drive a **20%** increase in ROI
- Constructed interactive **PowerBI** dashboards and **SQL** queries to streamline data access for business teams, reducing data request time by up to **90%** and enabling faster, data-driven decision-making
- Collaborated with the Advanced Analytics team to implement the **North Star Metric**, enhancing customer targeting accuracy by at least **10%** for promotions and improving overall campaign performance

Full Stack Developer

Sep. 2023 – Dec. 2023

SparkLease Inc.

Toronto, ON

- Designed and developed the user profile feature in WeChat Mini App using **React Native**, **XML**, **CSS**, and **JavaScript**, enabling seamless post management and optimizing process speed by **50%**
- Engineered responsive PC and mobile websites with **ASP.NET**, **CSHTML**, **C#**, and **JavaScript**, enhancing user navigation by **30%** through a dynamic filter system and improved account management
- Created a recommendation algorithm using **C#** and **SQL** on Microsoft Azure, increasing data retrieval speed by **30%** and improving recommendation accuracy by **15%**, leading to higher user satisfaction

PROJECTS

- NBA Win Probability Simulator** | *Python, XGBoost, Scikit-Learn, Pandas, SQLite* Mar. 2026 - Apr. 2026
- Build a comprehensive and scalable ML pipeline using **XGBoost** and **Logistic Regression** to predict NBA game outcomes across 5 seasons, achieving **66.9%** accuracy and outperforming the baseline predictor by **21%**
 - Engineered 73 complex game and player performance features using **Pandas** and **Scikit-Learn**, and implemented **SQLite** caching to efficiently process 5,400+ API requests without hitting rate limits